

## **MSE 546 Project Proposal**

Air Quality Data in India (2015 - 2020)

<https://www.kaggle.com/datasets/rohanrao/air-quality-data-in-india/data>

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## Project Identification & Task

The dataset that we are working with is the [Air Quality Data in India](#) Kaggle competition. From the initial competition description, it's shown that the datasets cover 26 cities and 230 stations that range from January 2015 to July 2020 on both daily and hourly levels. This proposal will focus on the "city\_data.csv" dataset. There are 12 features that contain the amount of each chemical in a cubic meter of air and an Air Quality Index (AQI). The task for this competition is regression. More specifically, the task to focus on is to predict the daily AQI given a city with its chemical levels, time of year, and historical AQI readings.

## Feasibility

The "dataset contains around 25,000 rows after dropping rows with unknown or unusable AQIs, and the data is a time series set, aggregated by day. The dataset can ultimately be used for both classical machine learning algorithms and neural network-based models. Classical ML algorithms such as Random Forest, Ridge and Gradient Boosting can be trained on a local CPU, whereas neural network models such as 1-D CNN, LSTM can be trained on Google Colab to provide sufficient processing power.

## Exploratory Data Analysis Insights

Within our initial data analysis, we highlighted four distinct data insights that helped to inform our modelling strategy. First, as shown in Figure 1a, the target (AQI) is right-skewed (skewness  $\approx 3.40$ ) with a mean of 166 as compared to its median of 118. We intend to log-transform the target to prevent our models from being too biased by extreme outliers. Second, as seen in Figure 1b, we found a strong season signal where winter AQI levels (221) are nearly double those of the monsoon season (116), indicating that temporal features like 'month' or 'season' are critical predictors for the model, potentially capturing unobserved seasonal factors like weather patterns. Third, as seen in Figure 1c, while pollutants (like PM2.5) display strong linear correlations with the target, our VIF analysis showed that multicollinearity is not severe enough to warrant feature removal. Finally, we observed significant variation across the 26 monitored cities, where industrial hubs showed distinct pollution baselines compared to the coastal regions. We will account for these location-specific differences using one-hot encoded city features.

## Evaluation Metrics

We used the following three metrics to evaluate our data. The values use the original AQI scale after inverting the log transform for computation. We used root mean square error as a metric to penalize large errors on high pollution days. Mean absolute error was used to find the average absolute AQI error in index units to measure prediction error. Additionally, we separated MAE by AQI category (Good Severe) to examine tail performance. We do this because underestimating severe pollution events carries the greatest real-world risk.  $R^2$  was used to give a proportion of AQI variance, providing a scale-free variance explanation that can be used to compare different models.

## Baseline Models & Performance

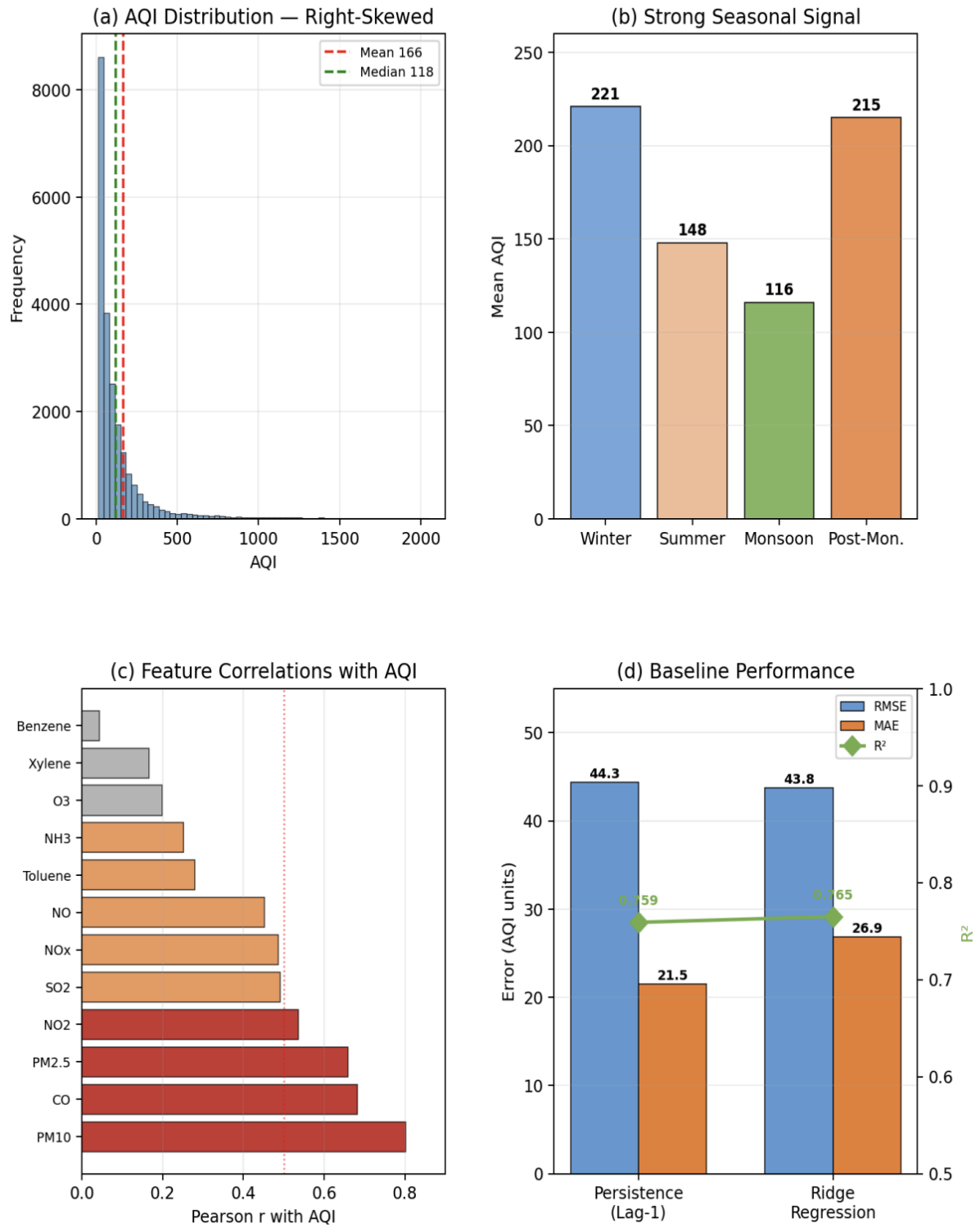
We implemented two baselines through a temporal train/test split (Cutoff 2019-12-01; 79.5% Train / 20.5% Test) to prevent future-data leaks. Rows with missing AQI values were dropped. Other missing features were supplied by medians for the respective city, and had flags to mark the row as having missing values.

- Persistence (Lag-1): Predict today's AQI as yesterday's value. This is a simple time-series baseline and sets a naive floor. Result - RMSE: 44.34 | MAE: 21.52 |  $R^2$ : 0.759.
- Ridge Regression ( $\alpha = 1.0$ ): A linear model trained on  $\log_{1+x}$  (AQI) with 59 features: 12 raw pollutants, 12 missingness flags, AQI lag-1 & lag-7, 7-day rolling mean & std, month, day-of-week, is\_weekend, 3 season dummies, and 25 city dummies. Result - RMSE: 43.77 | MAE: 26.85 |  $R^2$ : 0.765.

Ridge regression does slightly better on RMSE/ $R^2$ , but has higher MAE, showing that it captures more variance at the cost of over-smoothing extreme values. An example is Ridge's per-category MAE on Severe days. It's 201.5 vs. the overall 26.8, highlighting the need for nonlinear and temporal models that better capture extremes.

## Figures & References

Figure 1: Exploratory Data Analysis & Baseline Results



[1] Rohan Rao. Air Quality Data in India. Kaggle Datasets, 2020.

[kaggle.com/datasets/rohanrao/air-quality-data-in-india](https://kaggle.com/datasets/rohanrao/air-quality-data-in-india)